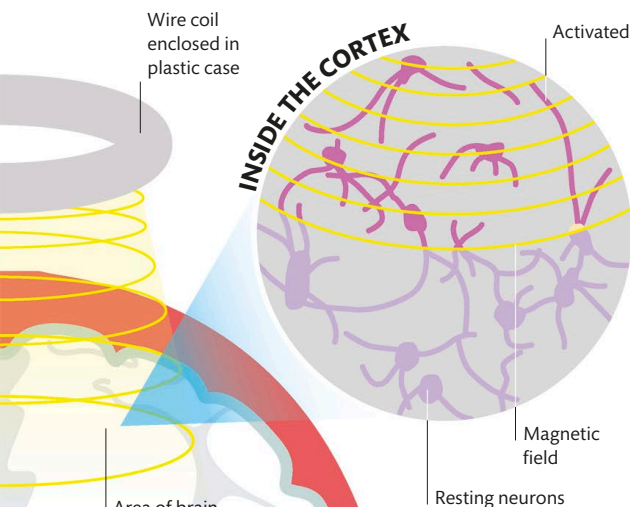


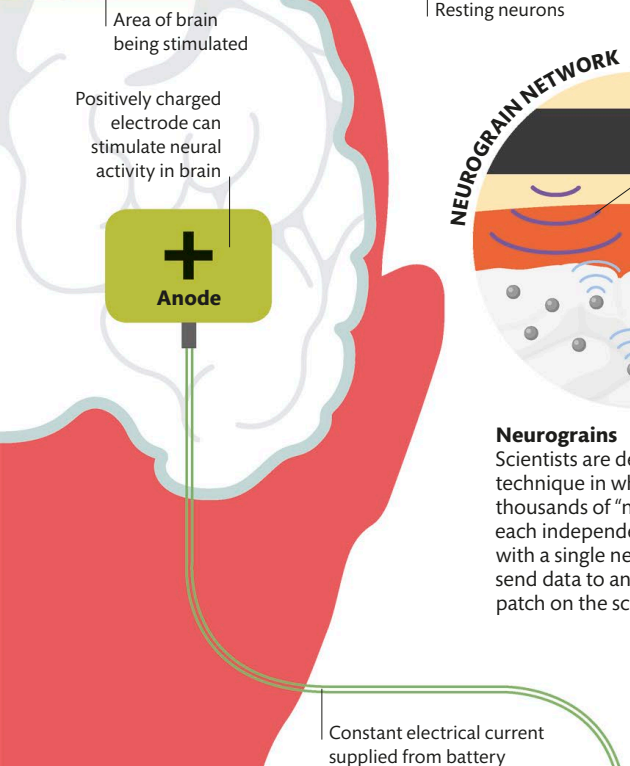


Transcranial magnetic stimulation

In transcranial magnetic stimulation (TMS), pulses of electric current pass through a coil and generate magnetism that penetrates the skull to influence brain cells and their impulses. The coil's position and motion, and pulse strength and timing, are adjusted to modify particular brain regions. TMS is being tried for many kinds of brain and behavioral conditions and also possibly to heighten thinking and other mental processes.

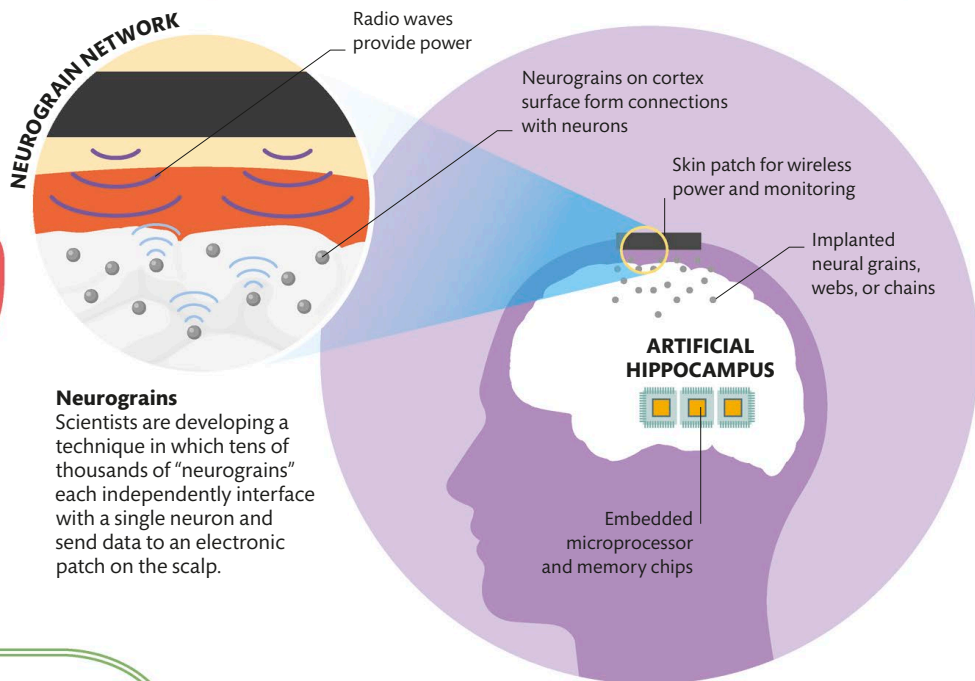
Magnetic pulse

When in use, the magnetic coils change polarity and produce magnetic pulses, which penetrate the scalp. This produces electrical activity in surrounding neurons.



Stimulating the brain

Anodal tDCS uses a positive current to speed up nerve cell activity. The positions of the skin electrodes determine which brain regions are aroused. Tests show that effects can persist even after the current ceases.



Neurograins

Scientists are developing a technique in which tens of thousands of "neurograins" each independently interface with a single neuron and send data to an electronic patch on the scalp.

Memory chips

The abilities of electronic devices can be extended by adding more memory, often in the form of microchips. The brain could be similarly upgraded. Microdevices to receive, store, and send data are being shaped like ultrafine webs, chains, and grains. Implanted on or in the cerebral cortex, they could develop connections with individual nerve cells and assist them in thinking and memory. Already, chips can advance hippocampus memory tasks such as long-term recall.